Kai Chen

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EDUCATION

University of Virginia Aug. 2024 – Now
Ph.D. in Computer Science Charlottesville, VA, USA

- **Research**: Data Synthesis (tabular/image/text generation); Generative AI; Machine Learning Algorithm; Differential Privacy; AI Security
- **Coursework**: Machine Learning; Reinforcement Learning; Cryptography; Computer Vision

University of Edinburgh Sep. 2020 – Aug. 2021
M.S. in Statistics with Data Science Edinburgh, UK

- **Coursework**: Generalized Regression Modeling; Large-scale Optimization; Machine Learning; Time Series

Zhejiang University Sep. 2016 – Jun. 2020
B.S. in Mathematics and Applied Mathematics & B.S. in Finance Hangzhou, CN

- **Coursework**: Mathematical Analysis; Linear Algebra; Probability Theory; Stochastic Process; Python Programming

PUBLICATION

- Kai Chen, Chen Gong, and Tianhao Wang. 2025. Beyond One-Size-Fits-All: Neural Networks for Differentially Private Tabular Data Synthesis. arXiv:2511.13893 [cs.LG] https://arxiv.org/abs/2511.13893
- Kai Chen, Xiaochen Li, Chen Gong, Ryan Mckenna, and Tianhao Wang. 2025. Benchmarking Differentially Private Tabular Data Synthesis: [Experiments & Analysis]. Proceedings of the ACM on Management of Data 3, 6 (Dec. 2025), 125. https://doi.org/10.1145/3769764
- Xiaoye Miao, Huanhuan Peng, Kai Chen, Yuchen Peng, Yunjun Gao, and Jianwei Yin. 2022. Maximizing time-aware welfare for mixed items. In 2022 IEEE 38th International Conference on Data Engineering (ICDE). IEEE, 10441057.

EXPERIENCE

ByteDance Dec. 2021 – May. 2023
Data Analyst Shanghai, CN

- Collect and process large-scale user data for subsequent analysis
- Identify key trends and inform business decisions through statistical analysis and modeling

PROJECTS

Differentially Private Multimodal Data Synthesis

- The first to focus on the DP multimodal data synthesis problem
- Propose a generic solution framework and design heuristic algorithms based on probabilistic modeling, diffusion model, and contrastive learning framework like CLIP, for the DP table-image multimodal data synthesis problem
- Systematically evaluate algorithms in this domain, proving the effectiveness of our proposed methods

Deep Learning Solution to Differentially Private Tabular Data Synthesis

- Analyze the strengths and weaknesses of statistical methods and NN theoretically and empirically
- Propose a new state-of-the-art neural network-based solution to DP tabular data synthesis by designing a distribution selection + model training pipeline
- Achieve state-of-the-art utility on multiple complex datasets, while improving speed compared to previous utility-optimal baselines

Differentially Private Tabular Data Synthesis Benchmark

- Propose a three-step framework for algorithm analysis
- Provide module-level comparisons on existing synthesis algorithms, including those traditional statistical models, GAN-based models, and diffusion models
- Paper is accepted to SIGMOD 2026, with a publicly available GitHub repository

Multidimensional Influence Maximization Project

- Propose the mixed-type influence diffusion problems in the social network
- Provide a solution to this problem with a theoretical guarantee and empirical superiority
- Paper is accepted to ICDE 2022

SKILLS

Programming : Python; PyTorch; JAX; R; SQL; Matlab; C
Languages : Native speaker of Mandarin; proficient in English